

MIDDLE EAST TECHNICAL UNIVERSITY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

EE201 – Circuit Theory I

Lecture 1: Basic Concepts

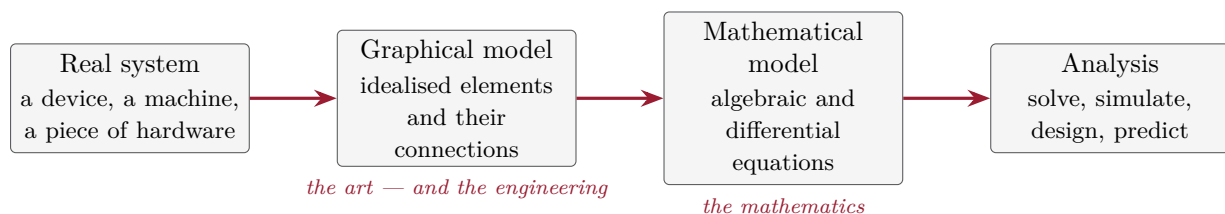
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Colour code used in these notes and in the slides

■ charge q ■ current i ■ voltage v ■ power p , energy w

1. Why circuits? From the world to a model

Circuit theory is not really about wires. It is a way of *modelling* physical systems, and you will use the same way of thinking in control, signal processing, power, robotics and electronics for the rest of your career. The pipeline is always the same:

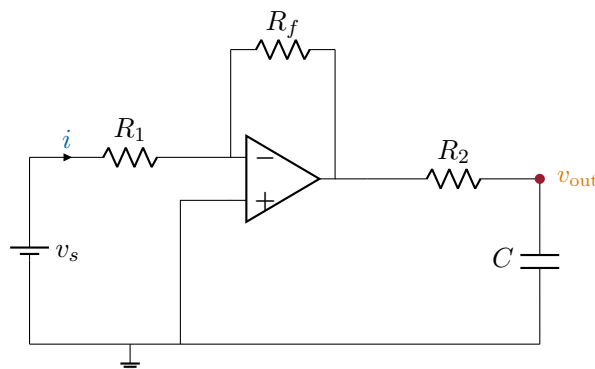


The hard step is the first arrow. A real resistor has inductance, a temperature coefficient, a power rating and a size; we throw all of that away and keep one number, R . Whether that is brilliant or foolish depends on what you are asking of the model. Calling that step “the art” makes it sound decorative, as though the equations were the real work. It is the other way round: the mathematics can be looked up, while deciding what to keep and what to discard *is* the engineering.

“All models are wrong, but some are useful.”

George E. P. Box, *Science and Statistics*, 1976

Every element you will meet in EE201 is wrong in some measurable way. The question is never “is this model true?” but “is it useful for what I am about to do?”.



A circuit of the kind you will be able to analyse by the end of the semester: a source, resistors, an operational amplifier and a capacitor — one element from each part of the course. All of EE201 is about what each element does on its own, and what the wires force them to share.

The same pipeline, other domains

The elements change, the structure does not. Every such domain has one *across* quantity and one *through* quantity:

Domain	“Across” quantity	“Through” quantity
electrical	electric potential	current
hydraulic	pressure	volumetric flow rate
thermal	temperature	heat flow
mechanical	velocity	force

An across quantity is always a *difference* between two points, and in every domain the word “difference” is dropped. Sometimes for convenience: “the voltage across R_2 ” names no terminals, yet nobody is confused. More often because a *reference* was agreed on long ago — altitude from sea level, gauge pressure from atmospheric pressure (a flat tyre reads zero, not 101 kPa), temperature from absolute zero. A circuit has no universal reference, so we choose one and call it *ground*.

A through quantity is the opposite. It is measured *at* one place — a cross-section of a wire, a pipe, a shaft — and needs a direction, not a second point. It needs no reference either: zero current means the same to everyone, while zero volts is somebody’s decision.

All these domains obey the same conservation and connection laws, which is why circuit theory is taught in most engineering disciplines.

In EE201 we work with the *lumped circuit* model, which we make precise at the end of this lecture. The sections before it build the vocabulary that model is written in: charge, current, voltage, ground, power, and a sign convention that keeps the bookkeeping honest.

2. Charge

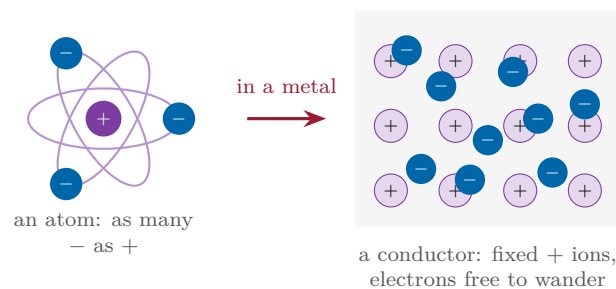
Electric charge

Charge is the physical property of matter that makes it experience a force in an electromagnetic field. It is denoted by q (or Q for a constant charge) and measured in **coulombs (C)**.

Charge is quantised: it comes in integer multiples of the elementary charge

$$e = 1.602 \times 10^{-19} \text{ C}, \quad \text{so} \quad 1 \text{ C} \approx 6.24 \times 10^{18} \text{ electron charges.}$$

An electron carries $-e$ and a proton $+e$. In an isolated atom the two exactly balance, so the atom as a whole is neutral.



Two facts about charge do all the work in this course:

1. **Charge is conserved.** It is neither created nor destroyed; it only moves. When we write Kirchhoff’s current law in Lecture 2, we are writing conservation of charge.

2. **A conductor is already full of charge, and is still neutral.** In a metal, roughly one electron per atom breaks loose and is free to move through the material, leaving a fixed positive ion behind. One cubic centimetre of copper contains about 8.5×10^{22} such free electrons — about 13 600 C of *mobile* charge — while the net charge of that copper is essentially zero, because every free electron is matched by the ion it left.

The second fact is easy to misread. A current in a wire is not charge being *poured into* the wire: the charge is already there, and a current is that charge *drifting*. The wire stays neutral while it drifts. This is the first assumption of the lumped model below, and it is what makes Kirchhoff's current law true.

3. Current

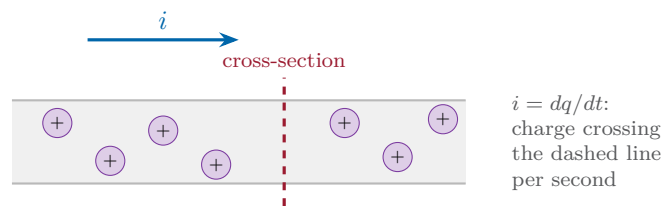
If charge sits still, nothing interesting happens. Circuits are about charge *in motion*.

Electric current

Current is the time rate of flow of charge through a cross-section:

$$i(t) = \frac{dq(t)}{dt} \quad \Longleftrightarrow \quad q(t) = q(t_0) + \int_{t_0}^t i(\tau) d\tau.$$

The unit is the **ampere (A)**, and $1 \text{ A} = 1 \text{ C/s}$.



Example 1.1 — a coulomb, and the speed of an electron

(a) **How much charge does a phone battery hold?** A 4000 mAh cell:

$$Q = 4 \text{ A} \times 3600 \text{ s} = 14\,400 \text{ C}, \quad N = \frac{14\,400}{1.602 \times 10^{-19}} \approx 9 \times 10^{22} \text{ electrons}.$$

Note that the rating is in ampere-hours — a charge written as a current times a time. And at 10^{22} charges the lumpiness of e is invisible, which is what lets us treat $q(t)$ as a smooth function and write dq/dt at all.

(b) **How fast do the electrons actually move?** Take 1 A flowing in an ordinary 1 mm^2 copper wire. Each electron drifts along the wire at roughly

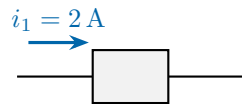
$$0.07 \text{ mm/s} \quad \text{— about one metre in four hours.}$$

Yet the lamp lights the instant you flip the switch. The carriers crawl; the *signal* that sets them all moving travels at nearly the speed of light.

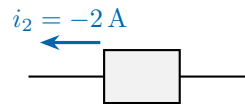
(For the curious: the drift speed follows from $v = i/(neA)$, where $n \approx 8.5 \times 10^{28} \text{ m}^{-3}$ is the density of free electrons in copper. You will not need this formula in EE201 — it is here so that the number is not magic.)

3.1. Reference direction: the arrow is a choice, not a fact

A current value is meaningless without an arrow. The arrow declares *which way we call positive*; the sign of the number then tells us which way the charge actually moves.



charge flows to the right



exactly the same physical situation

Both pictures describe the same physical current. You will often assign arrows before you know the answer; a negative result is not a mistake, it is the model telling you the charge goes the other way.

Conventional current

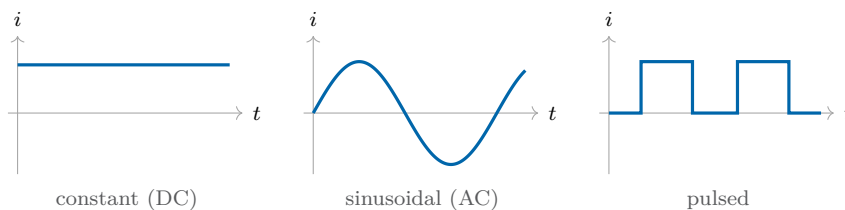
By historical convention, the current arrow points in the direction in which *positive* charge would move. In a copper wire the carriers are electrons, so they actually move *against* the arrow.

No equation in this course changes because of it. A current counts only the *net rate at which charge crosses a section*, and an electron moving left carries as much charge to the right as a proton moving right. The model never asks what the carriers are — and it had better not, since in a semiconductor they are electrons *and* holes, and in a battery they are ions.

3.2. Notation

We follow the usual convention of the course: lowercase letters for quantities that may vary with time, uppercase for constants.

$$i(t), v(t), q(t), p(t) \quad (\text{general}) \qquad I, V, Q, P \quad (\text{constant, “DC”})$$



Example 1.2 — charging a phone

In the last part of a charge cycle the battery is topped up: the charge *stored in the battery* approaches its full value as

$$q(t) = Q_f (1 - e^{-t/\tau}), \quad Q_f = 3600 \text{ C} (= 1000 \text{ mAh}), \quad \tau = 20 \text{ min} = 1200 \text{ s}.$$

The charging current is the rate at which that charge accumulates:

$$i(t) = \frac{dq}{dt} = \frac{Q_f}{\tau} e^{-t/\tau} = \frac{3600}{1200} e^{-t/\tau} = 3 e^{-t/\tau} \text{ A}.$$

So the current starts at 3 A and decays: after 20 minutes it is $3/e \approx 1.1 \text{ A}$, after an hour about 0.15 A. The fuller the battery, the more slowly it fills — which is why the last ten per cent always takes so long.

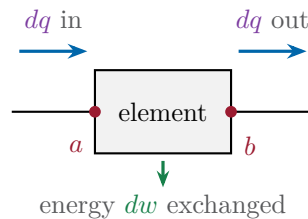
Going the other way, with a *constant* current the charge simply accumulates linearly: $I = 3 \text{ A}$ flowing for 8 hours transports $Q = I \Delta t = 3 \times 8 \times 3600 = 86\,400 \text{ C}$.

Charge is what accumulates; **current** is how fast it accumulates. Every integral–derivative pair in this course has that same flavour.

4. Voltage

Current tells us that charge is moving, but not *why* it moves or what the motion costs. Charge does not wander through a circuit on its own: carrying it through an element always exchanges energy, and voltage is the bookkeeping of that exchange.

Consider a small charge dq entering an element at terminal a and leaving at b , and let dw be the energy exchanged while that happens.



Voltage (potential difference)

The **voltage** v_{ab} between terminals a and b is the energy exchanged *per unit of charge* carried from a to b through the element:

$$v_{ab} = \frac{dw}{dq}$$

The unit is the **volt (V)**, and $1 \text{ V} = 1 \text{ J/C}$. A positive v_{ab} means the charge *gives up* energy to the element on its way from a to b .

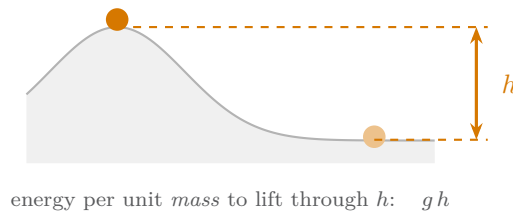
Four remarks, and they matter more than the formula.

1. **It is a ratio.** Carry twice the charge and twice the energy is exchanged, so v_{ab} comes out the same. Voltage therefore describes the *pair of terminals*, not the charge that happened to pass between them.
2. **Nothing has to be moving.** A 1.5 V battery in a drawer has 1.5 V across its terminals with $i = 0$, just as a closed tap still has pressure behind it. Voltage is a readiness to exchange energy, not a record of energy already exchanged.
3. **It is always between two points, and the order matters:** $v_{ab} = -v_{ba}$. “The voltage at node a ” means nothing until a reference point is chosen — which is why the older name *potential difference* is the more honest one.
4. **In a lumped circuit it depends only on the two endpoints, not on the path between them.** This is the second lumped-model assumption in disguise, and it is what makes “the voltage between a and b ” well defined. Applied around a closed path, it is Kirchhoff’s voltage law.

One thing comes for free. With $dq = i dt$, the energy exchanged per unit time is $dw/dt = v i$ — the power, which we never postulate: it follows from these two definitions.

4.1. Voltage as an electrical height

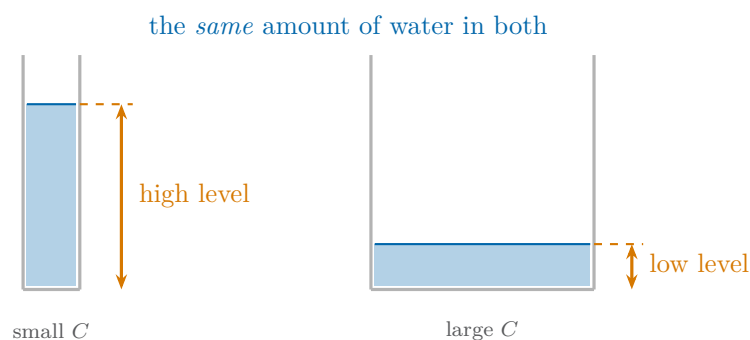
The most useful mental picture is gravitational. Lifting a mass m through a height h costs mgh ; the energy *per unit mass*, gh , depends only on the height difference. A heavier ball gains more energy, but energy per kilogram is the same — so gh , like voltage, belongs to the two places and not to the object you carried between them.



The picture anticipates two later ideas. “Sea level” is a choice, and moving it changes no height difference — that is ground. And the energy spent climbing a hill does not depend on the route — that is remark 4, and it is Kirchhoff’s voltage law.

4.2. Charge is the amount, voltage is the level

A trap that catches almost everybody: charge starts to *feel* like energy. It is not. The cleanest way to see why is to separate the *amount* of something from its *level*.



Pour one litre into a narrow tube and the level jumps; pour the same litre into a wide tank and it barely moves. Same amount, different level, because the two containers have different cross-sections. Electrically, the “cross-section” of a pair of conductors is called its **capacitance** C , and the relation is exactly the same one:

$$q = C v.$$

The *shape of the circuit* plays the part of the shape of the tank. The same charge moved between a different pair of points gives a different voltage, because the geometry — and therefore C — is different. Capacitance is defined properly in Lecture 5; for now it is enough that the link between *how much charge* and *what voltage* is a property of the circuit, not a universal constant.

Why charge cannot be energy in disguise

Charge the pair of conductors up to q . The energy you had to supply is

$$W = \int_0^q v dq' = \int_0^q \frac{q'}{C} dq' = \frac{q^2}{2C} = \frac{1}{2} C v^2.$$

Two things follow. First, energy grows with the *square* of charge — if charge were energy under another name, the relation would be linear. Second, look at the integral: the *first* coulomb costs almost nothing, because the level is still near zero, while the *last* coulomb is expensive, because by then it has to be lifted to the full height. The price of one coulomb *is* the voltage, and the voltage changes as you fill.

The sharpest statement of the difference is about a complete circuit:

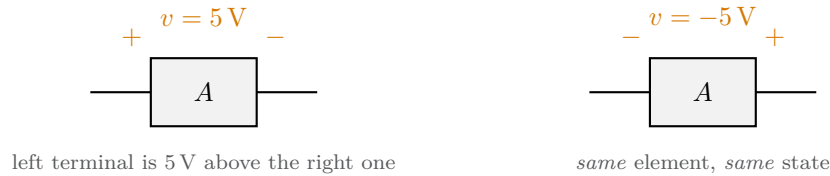
Around a loop, the charge that leaves the source comes back to it. The energy does not.

Charge *circulates*: every coulomb entering a resistor leaves it, and none is “used up”. Energy flows one way — out of the source, into heat, light or motion, and never back. The copper of the charge

section makes the same point: one cubic centimetre holds some 13 600 C of mobile charge and stores no energy at all. Charge is everywhere; what carries energy is the *difference in level*.

4.3. Reference polarity

Exactly as for current, we mark a *reference polarity* with a + and a – sign before we know the answer. The pair of pictures below describe the same physical element in the same physical state.



4.4. A first element: the resistor

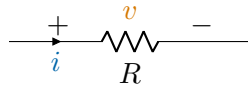
So far we have defined only *variables*: charge, current, voltage. A circuit also needs *elements*, each imposing a rule between the voltage across it and the current through it. That rule is the element's **branch relation**; the whole family arrives in Lecture 2. One of them is worth meeting now, because the analogies below need it.

The resistor and Ohm's law

A **resistor** is a two-terminal element whose voltage and current obey, at every instant, the algebraic relation

$$v = Ri$$

where R is the **resistance**, measured in **ohms** (Ω), with $1 \Omega = 1 \text{ V/A}$.



Two features of this relation matter more than the formula.

- **It is algebraic, and has no memory.** The current now is fixed by the voltage now — no derivatives, no integrals, nothing about the past. Such elements are called *resistive*, and the first four topics of the course live entirely among them. Capacitors and inductors, which *do* remember, arrive in Lecture 5 and change the problem completely.
- **The resistance comes from geometry and material.** For a uniform bar of length L and cross-section A made of a material of resistivity ρ ,

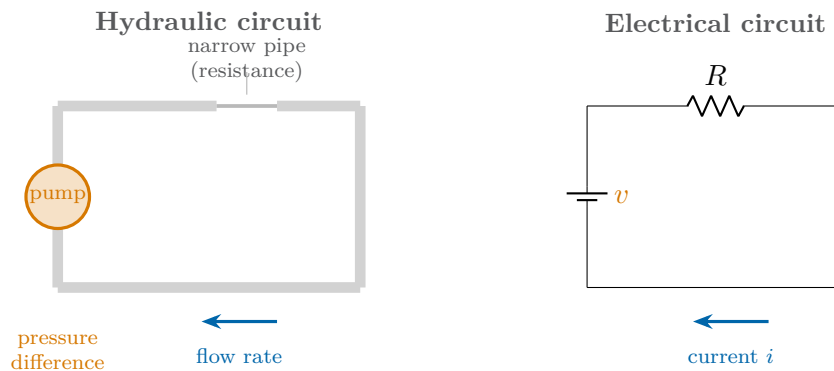
$$R = \rho \frac{L}{A}.$$

Long and thin means large R ; short and fat means small R . That is precisely the behaviour of a pipe carrying water, which is why the analogy of the next subsection works as well as it does. (Copper has $\rho \approx 1.7 \times 10^{-8} \Omega \cdot \text{m}$, which is why we make wires out of it and assume they have no resistance at all.)

We come back to the sign convention that accompanies $v = Ri$, and to the reason a resistor can only *absorb* energy, once power is defined.

4.5. The hydraulic analogy

Most people have better intuition for water than for electrons, and the analogy is close enough to be genuinely useful — as long as you remember it is an analogy.



Hydraulic	Electrical	Comment
pressure (difference)	voltage v	the “across” quantity: a difference between two points
volumetric flow rate	current i	the “through” quantity: needs a direction
pipe / tap restriction	resistance R	opposes flow, dissipates energy
pump	voltage source	supplies energy
water in the pipes	charge q	conserved: it does not disappear

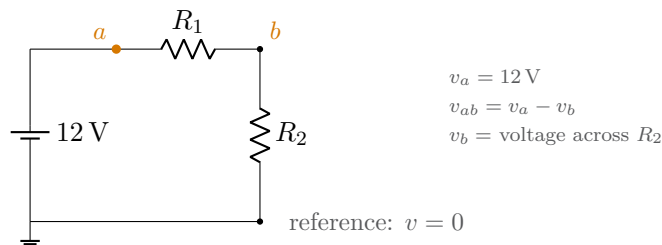
5. Ground and node voltages

Since only *differences* of potential have physical meaning, we are free to declare one point of the circuit to be zero. That point is called the **reference node**, or simply **ground**.

Ground (reference node)

A node chosen, by us, to have potential zero. Once it is chosen, the “voltage at node a ”, written v_a , means $v_{a,\text{ground}}$, and every two-point voltage follows from the difference

$$v_{ab} = v_a - v_b$$



Choosing a different node as ground shifts *every* node voltage by the same constant and changes *no* two-point voltage, no current and no power. The choice is a matter of convenience: pick the node that the most elements touch, and your equations get shorter.

Three different things are called “ground”

Earth ground is a physical connection to the planet, used for safety. **Chassis ground** is the metal enclosure of an instrument. **Signal ground** is the analysis reference we have just defined. In EE201 we mean the third one; in the laboratory, confusing them is how you blow a fuse.

6. Power and energy

Power and energy

Power is the rate at which energy is transferred:

$$p(t) = \frac{dw}{dt} = \frac{dw}{dq} \cdot \frac{dq}{dt} = v(t) i(t)$$

The unit is the **watt (W)**, $1 \text{ W} = 1 \text{ J/s}$. The energy transferred between t_0 and t is

$$w(t) = \int_{t_0}^t p(\tau) d\tau = \int_{t_0}^t v(\tau) i(\tau) d\tau.$$

Notice that $p = vi$ was not postulated: it fell out of the definitions of voltage (energy per charge) and current (charge per time). That is the whole point of defining them the way we did.

For the one element we have met so far, substituting Ohm's law gives

$$p = vi = i^2 R = \frac{v^2}{R},$$

which for $R > 0$ is never negative, whatever the signs of v and i . A resistor can only *absorb* energy — it has no way of giving any back — and that absorbed energy leaves the circuit as heat.

Energy bills are written in kilowatt-hours: $1 \text{ kWh} = 10^3 \times 3600 = 3.6 \times 10^6 \text{ J}$.

As a first check of the units: a 12 V battery supplying a constant 2 A for 5 minutes delivers

$$P = VI = 12 \times 2 = 24 \text{ W}, \quad W = P \Delta t = 24 \times 300 = 7200 \text{ J} = 7.2 \text{ kJ} = 2 \times 10^{-3} \text{ kWh}.$$

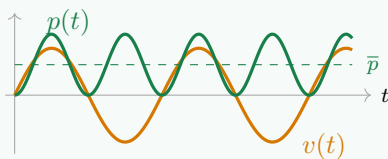
Example 1.3 — power that changes in time

A resistive element carries

$$v(t) = 10 \cos(\omega t) \text{ V}, \quad i(t) = 2 \cos(\omega t) \text{ A}, \quad \omega = 2\pi \cdot 50 \text{ rad/s}.$$

The instantaneous power is

$$p(t) = v(t) i(t) = 20 \cos^2(\omega t) = 10(1 + \cos 2\omega t) \text{ W}.$$



- $p(t) \geq 0$ always: a resistor never returns energy.
- The power ripples at *twice* the frequency of the voltage, about an average of 10 W.
- The *peak*, 20 W, is twice the average. An element must survive the peak; how hot it gets is set by the average.

The energy absorbed over one period $T = 1/50 \text{ s}$ is

$$w = \int_0^T p(t) dt = 10T + \underbrace{\int_0^T 10 \cos 2\omega t dt}_{=0} = \frac{10}{50} = 0.2 \text{ J}.$$

The average power is what a wattmeter shows and what heats the element; the 100 Hz ripple is what makes a cheap transformer hum.

7. The passive sign convention

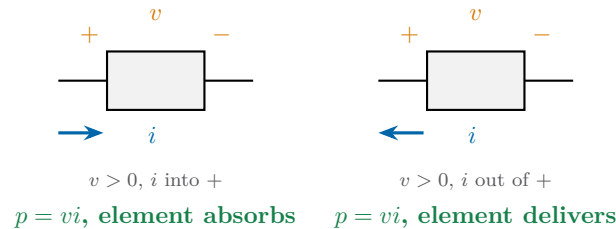
We now have $p = vi$, but v and i each carry a reference that *we* chose. Which sign of p means “this element is consuming energy”? A convention settles it once and for all.

Passive sign convention (PSC)

Assign the current reference arrow so that it **enters the terminal marked +**. With that choice,

$$p = vi = \text{power absorbed by the element}$$

so $p > 0$ means the element absorbs (consumes) energy, and $p < 0$ means it delivers (supplies) energy to the rest of the circuit.



If the picture you are given does *not* satisfy the PSC — the arrow leaves the + terminal — there is no need for a second formula. Flip one reference, change that variable's sign, and use $p = vi$ as always:

$$p_{\text{absorbed}} = v(-i) = -vi.$$

7.1. Conservation of power

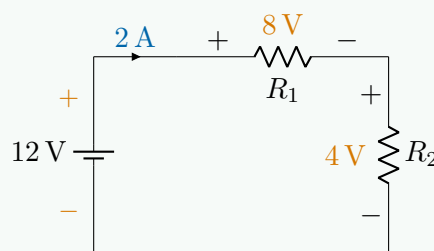
Energy is conserved, so in *any* circuit, at *every* instant, the absorbed powers of all elements sum to zero:

$$\sum_{k=1}^N p_k(t) = 0$$

Elements with $p_k < 0$ (sources, usually) supply exactly what the elements with $p_k > 0$ (resistors, usually) consume. This is Tellegen's theorem. If your powers do not cancel, something is wrong.

Example 1.4 — a power balance

In the circuit below the source drives 2 A around the loop and the two resistors drop 8 V and 4 V.



With the PSC applied to each element:

$$p_{R_1} = 8 \times 2 = +16 \text{ W}, \quad p_{R_2} = 4 \times 2 = +8 \text{ W}, \quad p_{\text{source}} = 12 \times (-2) = -24 \text{ W}.$$

The source current leaves its + terminal, hence the -2 A ; the sum is $16 + 8 - 24 = 0$, as it must be. The source *delivers* 24 W and the resistors *absorb* it.

8. Electric circuits and the lumped model

Electric circuit

An **electric circuit** is an interconnection of electrical elements through which charge can flow along closed paths.

Everything we do in EE201 rests on one further modelling assumption, and it deserves to be stated precisely rather than assumed silently.

The lumped circuit model

A circuit is **lumped** when three things may be assumed.

(a) **Nothing piles up.** At every instant, the current leaving an element equals the current entering it.

(b) **Nothing leaks out.** No changing *magnetic* field passes through the loops formed by your wiring, and no charge on one element pulls on the charge in another across the space between them. The fields belong to the elements, not to the space between them.

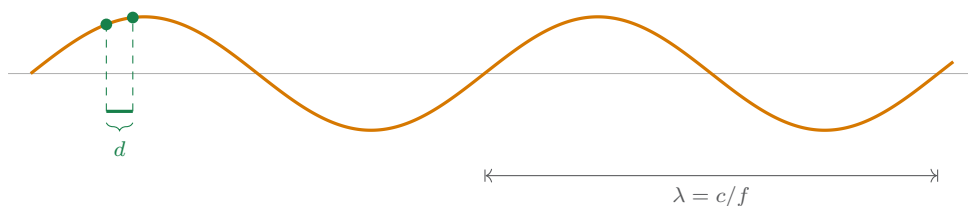
(c) **Nothing is late.** A signal of frequency f travelling at the speed of light c advances a distance $\lambda = c/f$ in one period; λ is its **wavelength**. The circuit, of size d , must be much smaller than that:

$$d \ll \lambda = \frac{c}{f}, \quad c \approx 3 \times 10^8 \text{ m/s.}$$

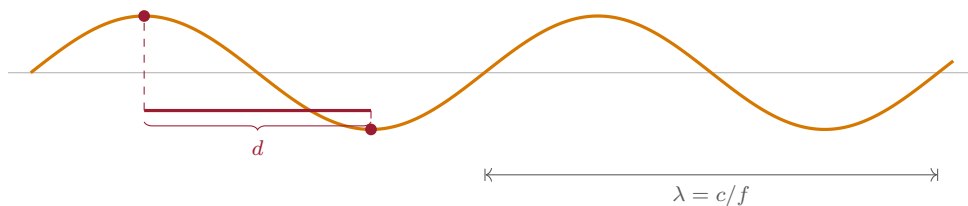
Read (c) in terms of times if you prefer: crossing the circuit takes d/c , while the signal itself changes in $1/f$, and we assume the first is negligible next to the second. Either way, a change at one point reaches everywhere else before anything else has had time to happen.

When the three hold, a circuit is described by *ordinary* differential equations in time. When they do not, the same physics needs Maxwell's equations and partial differential equations in space and time — which is what the electromagnetic field theory courses are for.

LUMPED: $d \ll \lambda$ both ends of the circuit sit at practically the same point of the wave



NOT LUMPED: $d \sim \lambda$ one end is at the crest while the other is at the trough



a *snapshot* along the circuit at one instant — the horizontal axis is distance, not time

The picture above is the one to keep. It shows the signal *frozen in space*: the horizontal axis runs along the circuit, and the curve is the value the signal has at each place at one instant. If the circuit occupies a tiny fraction of a wavelength, every point of it reads the same number, and we may speak of “the” voltage of a node. If it occupies an appreciable fraction, different parts of the same circuit hold different values at the same moment, and a single node voltage no longer exists.

Each of the three deserves a sentence of translation.

- (a) **does not forbid charge from moving *inside* an element.** A capacitor is the obvious case: it puts $+q$ on one plate and $-q$ on the other, so charge certainly separates inside it. What (a) says is that the element as a *whole* never gains or loses charge — the same current $i = dq/dt$ that enters one terminal leaves the other. A wire, likewise, is a pipe and not a bucket.

Without this, “the current through an element” would not even be a well-defined phrase, and Kirchhoff’s current law would be false.

- **(b) says the loops in your diagram are not antennas and not transformers.** You drew a rectangle of wire; if a changing *magnetic* field passed through that rectangle it would induce a voltage in it, and the voltage between two nodes would depend on how you routed the wire. Assumption (b) rules that out, which is what makes “the voltage between *a* and *b*” independent of the path — and Kirchhoff’s voltage law true.
- **(c) says everything happens everywhere at once.** A signal crosses a 30 cm board in about a nanosecond, while a 50 Hz mains waveform takes milliseconds to change appreciably — a factor of a million. That delay is not worth modelling, so we may use a single time variable *t* for the whole circuit instead of a function of position as well.

LUMPED

mains, $f = 50 \text{ Hz} \Rightarrow \lambda = 6000 \text{ km}$
 a 30 cm board: $d/\lambda \sim 10^{-7}$
 — circuit theory applies

NOT LUMPED

Wi-Fi, $f = 2.4 \text{ GHz} \Rightarrow \lambda = 12.5 \text{ cm}$
 a 3 cm trace: $d/\lambda \sim 1/4$
 — you need field theory

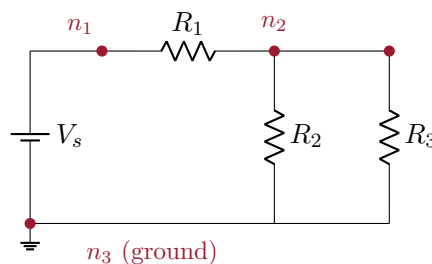
The same piece of hardware can be lumped for one signal and distributed for another. A mobile phone motherboard is a lumped circuit as far as its battery is concerned and a transmission-line problem as far as its antenna is concerned. The model is chosen by the question, not by the object.

9. Terminals, nodes, branches

The last piece of vocabulary describes how elements are connected — the topology.

Terminal, node, branch, loop

- A **terminal** is a point at which an element can be connected to others. A resistor, capacitor or inductor has two terminals; a transistor has three; an op-amp has five or more.
- A **node** is a point of connection between two or more elements. Everything joined by ideal (zero-resistance) wire is *one and the same node*, however long the wire is drawn.
- A **branch** is a single element between two nodes.
- A **loop** is a closed path through the circuit that does not pass through any node more than once.

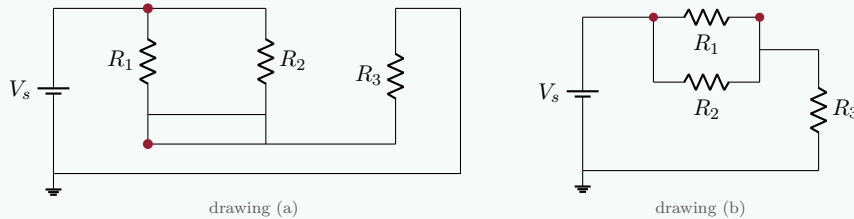


In the figure, n_2 is a single node even though three elements meet there through a stretch of wire; the wire itself is not a branch. Counting: 3 nodes, 4 branches (V_s , R_1 , R_2 , R_3) and 2 independent loops, consistent with the relation you will meet in Lecture 2,

$$\# \text{branches} = \# \text{loops} + \# \text{nodes} - 1 \quad \Rightarrow \quad 4 = 2 + 3 - 1.$$

Example 1.5 — the same circuit, drawn twice

The two drawings below look different but are the *same* circuit: in both, R_1 and R_2 have both of their terminals in common, and that pair is in series with R_3 and the source.



Drawing (b) makes it obvious that R_1 and R_2 are in parallel. Drawing (a) hides it behind a detour of wire. Nothing was changed except where the symbols were placed — and that is the only skill being tested when a problem looks impossible at first sight.

Drawing is not topology

Two circuits that look completely different on paper can be the same circuit. What matters is which terminals share a node, not where you placed the symbols. So when a problem looks unfamiliar, redraw it: put every element that touches a given node so that it visibly touches the *same point or the same line* as the others on that node, and straighten out the detours of wire. Series and parallel connections usually become obvious the moment you do this, and half of the difficulty disappears with them.

Summary

Quantity	Symbol	Defining relation	Unit
charge	q	—	coulomb, C
current	i	$i = dq/dt$	ampere, A = C/s
voltage	v	$v = dw/dq$	volt, V = J/C
power	p	$p = dw/dt = vi$	watt, W = J/s
energy	w	$w = \int p dt$	joule, J

- Current needs a **reference arrow**; voltage needs a **reference polarity**. Both are choices. A negative answer is information, not an error.
- With the **passive sign convention** (current into the + terminal), $p = vi$ is the absorbed power, and $\sum_k p_k = 0$ over the whole circuit.
- Voltage is an **across** quantity, current is a **through** quantity; ground is the node we *choose* to call zero.
- The **lumped** assumption ($d \ll \lambda$, no charge accumulation, no stray flux) is what lets us use ordinary differential equations for the rest of the course.

Next lecture

Interconnection laws — Kirchhoff's current and voltage laws — and the branch relations of the basic elements. Everything from Lecture 1 is used there without further comment.

Reading

- Alexander & Sadiku, *Fundamentals of Electric Circuits*, Chapter 1.
- Chua, Desoer & Kuh, *Linear and Nonlinear Circuits*, Chapter 1.
- Lecture notes and videos of Prof. Emre Tuna, users.metu.edu.tr/etuna/ee201.